MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Customer: **Mansourah Massarah Gold Project**
MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: **DESIGN CALCULATION**
3610 POX CONDITIONING, LEACHING & ADSORPTION
DESIGN OF TANK FOUNDATION

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D01	22.11.2023	AS BUILT	AVG	SJM	AG	

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

This document is to Prepare for design of **Pox Conditioning, leaching & Adsorption (3610)-** Tanks's Foundation. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel and Staad. For member strength calculation force values are directly taken from Load Data Provided & Output of Staad analysis.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Foundation and Load Plan (Doc. No. MD-216-3610-EG-ME-PLN- 1003 & 1004) attached in Appendix A. Also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5 CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

5.3 Foundations for Aboveground Tanks

Foundation shall be Earth Foundation with a Concrete Ringwall. The design calculations are prepared as per recommendation of standard API 650 - Annex B.

5.4 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Wall	50
	Raft	50

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5.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 450 kN/m^2 at 1.2m depth below EGL (ie. RL 911.50m) refer Table 5 - BH-11 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

6. DESIGN CALCULATION FOR FOUNDATION

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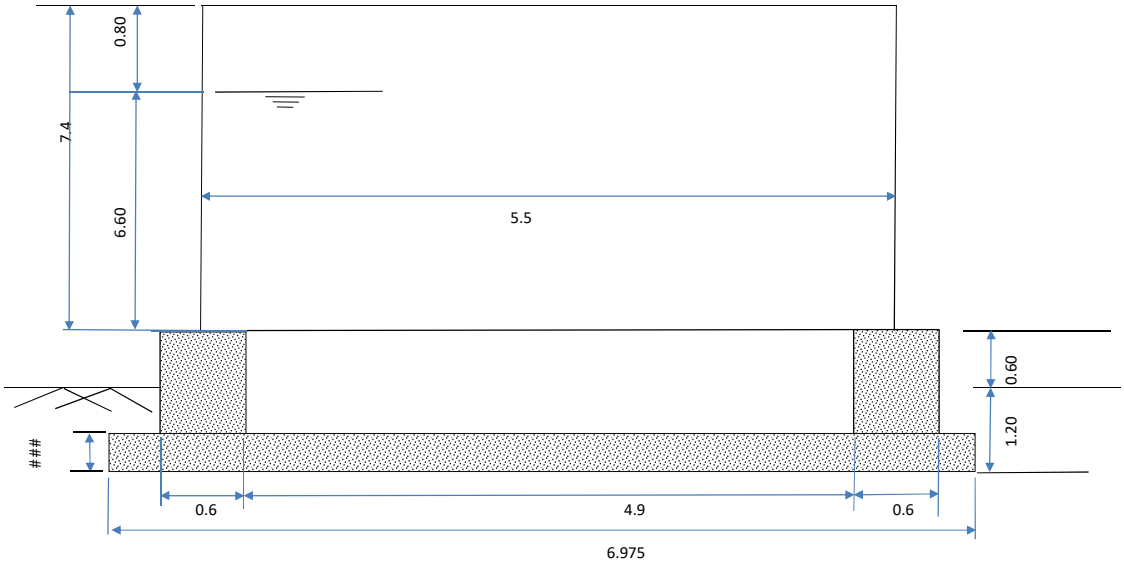
Foundation Marked F1

Tank Loading Data

DEAD WEIGHT OF EQUIPMENT	172.5	kN
DEAD WEIGHT OF CIVIL STRUCTURES	14	kN
OPERATIONAL WEIGHT OF SLURRY	2053	kN
MAXIMUM WEIGHT OF SLURRY	2302	kN
EXTERNAL PLATFORM LOAD	96.5	kN
WIND LOAD	48.7	kN
SEISMIC LOAD	98	kN
SEISMIC TORQUE AT FOUNDATION LEVEL	373	kNm

	Loading Type		
Weight, V	Empty (W)	283	kN
	Operating (Wo)	2336	kN
	Max (Wt)	2585	kN

Summary of Tank Specification



Diameter of Tank, D	=	5.5 m	
Height of Tank, H	=	7.4 m	
Height of Tank Fluid, H_L	=	6.6 m	
Inner Diameter of Concrete Ring Wall, ID	=	4.9 m	
Outer Diameter of Concrete Ring Wall, OD	=	6.1 m	
Height of Concrete Ring Wall, h	=	1.80 m	
Thickness of Concrete Ring Wall, t	=	0.6 m	
Thickness of Concrete Raft, T	=	0.50 m	
P.C.D of Bolts	=	5.772 m	
Top of Foundation	=	0.6 m	(Above Finished Ground Level)
Unit Weight of Liquid, γ_L	=	13.093 kN/m^3	
Unit Weight of Soil, γ_s	=	20 kN/m^3	
Unit Weight of Concrete, γ_c	=	24 kN/m^3	

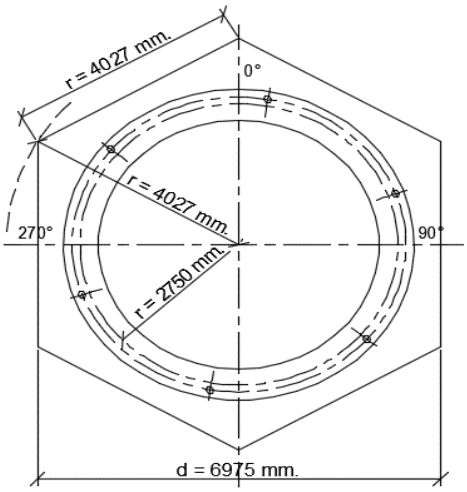
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Wind Load & Seismic Load

Total Base Shear due to wind load = 48.7 kN

Total Base Shear due to Seismic load = 98 kN

Load	Load Condition	Base Shear (F,kN)	Moment (M,kN-m)
Wind Loads	Empty	48.70	219.15
	Operating	48.70	219.15
Seismic Loads	Operating	98.00	373.00



Foundation Weight

a) Ring Wall Weight = 323.46 kN

b) Inside Soil Weight = 490.29 kN

c) Outside Soil Weight = 180.69 kN

d) Raft Weight = 505.57 kN

Area of Raft

Area of Raft (Hexagonal), $A = 2.598 * r^2$ = 42.13 m^2

Property of Hexagon

Radious of outer circle, r = 4.027 m

Dist. Of 2 opposite side, d = 6.975 m

Sectional Modulus of Raft

Sectional modulus (Hexagonal), $Ze = 0.0601*d^4 / r$ = 35.32 m^3

Basic Load Cases

1. Tank Dead Load (D)
2. Operational Fluid Load (F1)
3. Max Fluid Load (F2)
4. Wind Load (W)
5. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301 ,2007 Cl No. 2.4.1)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101	D + F1	2336.00	0.00	0.00
Short term	Empty	LC102	0.6 D + W	169.80	48.70	219.15
	Operating	LC103	D + F1 + W	2336.00	48.70	219.15
		LC104	D + F1+ 0.7 E	2336.00	68.60	261.10
		LC105	0.6 (D + F1) + 0.7 E	1401.60	68.60	261.10
	Maximun weight	LC106	D + F2	2585.00	0.00	0.00

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Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 350 KN/m^2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 465.5 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m^2	Qmax= V/A - M/Z kN/m^2	Qall/ Qmax	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	3836.01	0.00	91.05	91.05	3.84	OK
LC102	1669.81	306.81	48.32	30.95	9.63	OK
LC103	3836.01	306.81	99.74	82.36	4.67	OK
LC104	3836.01	384.58	101.94	80.16	4.57	OK
LC105	2901.61	384.58	79.76	57.98	5.84	OK
LC106	4085.01	0.00	96.96	96.96	3.61	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of V (kN)	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
LC101	3836.01	0.3	1150.80	0.00	-	-
LC102	1669.81	0.3	500.94	48.70	10.3	OK
LC103	3836.01	0.3	1150.80	48.70	23.6	OK
LC104	3836.01	0.3	1150.80	68.60	16.8	OK
LC105	2901.61	0.3	870.48	68.60	12.7	OK
LC106	4085.01	0.3	1225.50	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of V (kN)	Edge Dis.(d/2)	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
LC101	3836.01	3.4875	13378.10	0.00	-	-
LC102	1669.81	3.4875	5823.47	306.81	19.0	OK
LC103	3836.01	3.4875	13378.10	306.81	43.6	OK
LC104	3836.01	3.4875	13378.10	384.58	34.8	OK
LC105	2901.61	3.4875	10119.38	384.58	26.3	OK
LC106	4085.01	3.4875	14246.48	0.00	-	-

Ultimate Strength Design

Load Combination (As per SBC-304 ,2018 Table 5.3.1)

				Load at bottom of tank Inner Area			
Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC201	1.4 (D + F1)	3270.40	0.00	23.76	16.33
Short term	Empty	LC202	0.9 D + 1.6 W	254.70	350.64	23.76	16.33
	Operating	LC203	1.2 (D + F1) + 1.6 W	2803.20	350.64	23.76	16.33
		LC204	1.2 (D + F1) + E	2803.20	373.00	23.76	16.33
		LC205	0.9 (D + F1) + E	2102.40	373.00	23.76	16.33
	Test	LC206	1.4 (D + F2)	3619.00	0.00	23.76	16.33

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Staad input calculation.

Total Number of Noads in Wall Top = 36

S/W of tank (Empty Condition) on each Nodes= 7.861 kN

Pressure due to sand fill on inside Raft= 26.0 kN/m^2

Pressure due to soil fill on outside Raft= 14.0 kN/m^2

Pressure due to F1 on inside Raft= 86.5 kN/m^2

Pressure due to F2 on inside Raft= 96.9 kN/m^2

Pressure due to F1 on Wall= 43.228 kN/m^2

Pressure due to F2 on Wall= 48.471 kN/m^2

Wind Base Shear of tank on each Nodes= 1.353 kN

Seismic Base Shear of tank on each Nodes= 2.722 kN

Sl No.	Node No.	Dist. R(m)	R^2	Seismic V(kN)	Wind V(kN)
1	120	0.00	0.00	0.000	0.000
2	121	0.48	0.23	1.310	0.769
3	122	0.94	0.89	2.578	1.515
4	123	1.38	1.89	3.767	2.213
5	124	1.77	3.13	4.844	2.846
6	125	2.11	4.44	5.773	3.392
7	126	2.38	5.67	6.526	3.834
8	127	2.58	6.68	7.080	4.160
9	128	2.71	7.33	7.419	4.359
10	129	2.75	7.56	7.534	4.427
11	130	2.71	7.33	7.419	4.359
12	131	2.58	6.68	7.080	4.160
13	132	2.38	5.67	6.526	3.834
14	133	2.11	4.44	5.773	3.392
15	134	1.77	3.13	4.844	2.846
16	135	1.38	1.89	3.767	2.213
17	136	0.94	0.89	2.578	1.515
18	137	0.48	0.23	1.310	0.769
19	138	0.00	0.00	0.000	0.000
20	139	-0.48	0.23	-1.310	-0.769
21	140	-0.94	0.89	-2.578	-1.515
22	141	-1.38	1.89	-3.767	-2.213
23	142	-1.77	3.13	-4.844	-2.846
24	143	-2.11	4.44	-5.773	-3.392
25	144	-2.38	5.67	-6.526	-3.834
26	145	-2.58	6.68	-7.080	-4.160
27	146	-2.71	7.33	-7.419	-4.359
28	147	-2.75	7.56	-7.534	-4.427
29	148	-2.71	7.33	-7.419	-4.359
30	149	-2.58	6.68	-7.080	-4.160
31	150	-2.38	5.67	-6.526	-3.834
32	151	-2.11	4.44	-5.773	-3.392
33	152	-1.77	3.13	-4.844	-2.846
34	153	-1.38	1.89	-3.767	-2.213
35	154	-0.94	0.89	-2.578	-1.515
36	155	-0.48	0.23	-1.310	-0.769
		ΣR^2=	**136.14**		

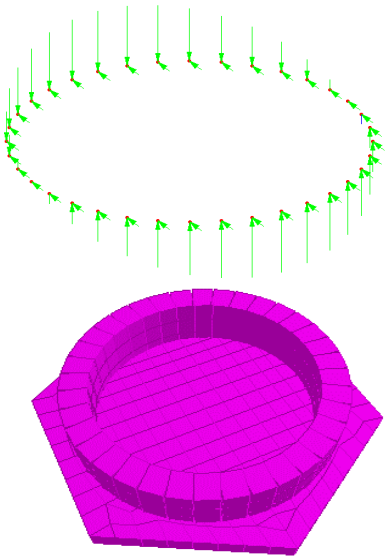
Staad input calculation For Seismic and Wind load distribution.

$V_{wind} = R * W_{total} / \Sigma R^2$ $H_{wind} = W_{total} / 36$

$V_{Seismic} = R * E_{total} / \Sigma R^2$ $H_{seismic} = E_{total} / 36$

H_{wind} = 1.353 kN

$H_{seismic}$ = 2.722 kN



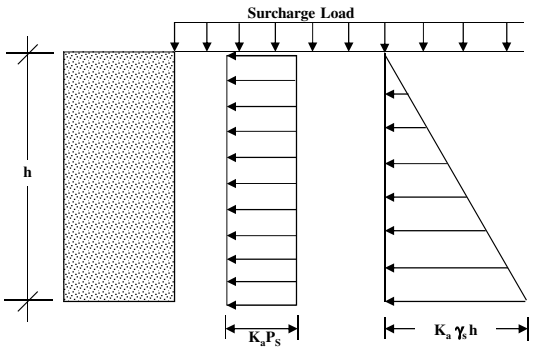
Staad Model in 3D view

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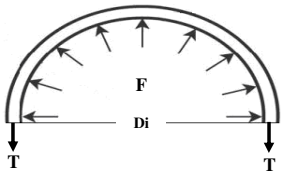
Wall Design

Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z
	V (kN)	M(kNm)	kN/m²
LC201	3270.40	0.00	137.65
LC202	254.70	350.64	32.19
LC203	2803.20	350.64	139.46
LC204	2803.20	373.00	140.82
LC205	2102.40	373.00	111.33
LC206	3619.00	0.00	152.33



Maximum Surcharge Load, Ps = 152.33 kN/m²



Where, K_a = coefficient of earth pressure at rest
= 0.5

Pressure due to Surcharge,P1 = $K_a * P_s$ = 76.16 KN/m²

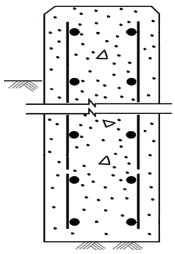
Pressure due to Soil,P2 = $1.4 * \gamma_S * (h-T) * K_a$ = 18.20 KN/m² for LC206 - 1.4 (D + F2)

Maximum Lateral Load

F = P1 * (h-T) + 1/2 * P2 *(h -T)= 110.8417 KN/m

Hoop Tension Load

T= 1/2 * F * ID = 271.5623 KN



Hoop Tension Bar

1. Horizontal Reinfocement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = T/(ϕ x Fy) = 1950 mm²

Minimum As = 0.0025 x t x h = 1950 mm² (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 14 Nos. D16= 2540.417 mm² OK

2. Vertical Reinfocement

Minimum As = 0.0015 * b * D = 900 mm² (As per API 650 , Cl. No. B.4.2.3)

STAAD output:- Wall Thickness = 600 mm

Characteristic strength of concrete, f'_c = 35 MPa

Yield stress of steel reinforcement, f_y = 420 MPa

Clear cover for main reinforcement bars = 50mm

Max BM (+) (M_y)=	1.9	kN m/m	(VERTICAL MOMENT IN INSIDE FACE)
Max BM (-) (M_y)=	59.05	kN m/m	(VERTICAL MOMENT IN OUTSIDE FACE)
Max BM (+) (M_x)=	0.3	kN m/m	(HORIZONTAL MOMENT IN INSIDE FACE)
Max BM (-) (M_x)=	10.39	kN m/m	(HORIZONTAL MOMENT IN OUTSIDE FACE)

Max Shear Qx =	0.01	N/mm²
Max Shear Qy =	0.17	N/mm²

Effective depth of wall in Vertical direction = 542 mm

Effective depth of wall in Horizontal direction = 526 mm

MA'ADEN معادن										Outotec / LARSEN & TOUBRO	
MD-216-3610-EG-CV-CAL-1304					Mansourah Massarah Gold Project						
900707_GAU01_HL12_CED01_ 36102_OU100672892_R3					DESIGN CALCULATION POX CONDITIONING, LEACHING & ADSORPTION DESIGN OF TANK FOUNDATION					Date: 22.11.2023	
O19045_450_C_K_24131_001_R3					AS BUILT					Rev: D01	
Moment	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
My(+)	1.90	0.01	0.01	0.000	11.13	975.60	D16	206.09	300	200	1005.31
My(-)	59.05	0.20	0.27	0.001	347.44	975.60	D16	206.09	300	200	1005.31
Mx(+)	0.30	0.00	0.00	0.000	1.81	975.60	D16	206.09	300	200	1005.31
Mx(-)	10.39	0.04	0.05	0.000	62.76	975.60	D16	206.09	300	200	1005.31

Check against one-way shear force

Allowable shear stress, (ϕ Vc)/bd = 0.74 N/mm^2

Shear stress at a distance 'd' from raft top , τ_v= 0.17 N/mm^2 **OK**

SUMMARY OF REINFORCEMENT IN WALL: **600 mm**

VERTICAL REIN. IN INSIDE FACE	**D16**	**200**	**c/c**	
VERTICAL REIN. IN OUTSIDE FACE	**D16**	**200**	**c/c**	
HORIZONTAL REIN. IN INSIDE FACE	**D16**	**200**	**c/c**	**(7Nos.)**
HORIZONTAL REIN. IN OUTSIDE FACE)	**D16**	**200**	**c/c**	**(7Nos.)**

Raft Design

STAAD output:- Raft Thickness = 500 mm

Max BM (+) (M_y)=	28.43	kN m/m	(HORIZONTAL MOMENT IN TOP FACE)
Max BM (-) (M_y)=	35.93	kN m/m	(HORIZONTAL MOMENT IN BOTTOM FACE)
Max BM (+) (M_x)=	14.13	kN m/m	(LONGITUDINAL MOMENT IN TOP FACE)
Max BM (-) (M_x)=	14.13	kN m/m	(LONGITUDINAL MOMENT IN BOTTOM FACE)

Max Shear Qx =	0.17	N/mm^2
Max Shear Qy =	0.18	N/mm^2

Effective depth of wall in Vertical direction = 442 mm

Effective depth of wall in Horizontal direction = 426 mm

Moment	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
My(+)	28.43	0.15	0.19	0.000	204.86	795.60	D16	252.72	300	250	804.25
My(-)	35.93	0.18	0.25	0.001	259.13	795.60	D16	252.72	300	250	804.25
Mx(+)	14.13	0.08	0.10	0.000	105.48	766.80	D16	262.21	300	250	804.25
Mx(-)	14.13	0.08	0.10	0.000	105.48	766.80	D16	262.21	300	250	804.25

Check against one-way shear force

Allowable shear stress, (ϕ Vc)/bd = 0.74 N/mm^2

Shear stress at a distance 'd' from raft top , τ_v= 0.18 N/mm^2 **OK**

SUMMARY OF REINFORCEMENT IN RAFT: **500 mm**

(HORIZONTAL REIN. IN TOP FACE)	**D16**	**250**	**c/c**
(HORIZONTAL REIN. IN BOTTOM FACE)	**D16**	**250**	**c/c**
(LONGITUDINAL REIN. IN TOP FACE)	**D16**	**250**	**c/c**
(LONGITUDINAL REIN. IN BOTTOM FACE)	**D16**	**250**	**c/c**

Software licensed to LNT ECC DIV	Job No O19045	Sheet No 1 / Rev
	Part	
Job Title Mansourah Massarah Gold Project	Ref	
	By AvG / Date 12-Feb-20 / Chd SJM	
Client Ma'aden	File POX CIL (3610).std	Date/Time 25-Feb-2020 10:45

Plate Center Stress Summary

			Shear		Membrane			Bending		
	Plate	L/C	Qx (N/mm^2)	Qy (N/mm^2)	Sx (N/mm^2)	Sy (N/mm^2)	Sxy (N/mm^2)	Mx (kN^-m/m)	My (kN^-m/m)	Mxy (kN^-m/m)
Max Qx	141	206:1.4 (D + F	**0.17**	-0.05	0.00	0.00	0.00	-31.52	-14.12	8.40
Min Qx	311	206:1.4 (D + F	**-0.17**	-0.05	0.00	0.00	0.00	-31.52	-14.12	-8.40
Max Qy	231	206:1.4 (D + F	0.01	**0.18**	0.00	0.00	0.00	-10.53	-35.77	2.45
Min Qy	133	206:1.4 (D + F	0.04	**-0.17**	0.00	0.00	0.00	-10.90	-35.93	3.44
Max Sx	126	201:1.4 (D + F	-0.04	-0.03	**0.00**	0.00	0.00	-3.77	-2.54	-2.47
Min Sx	126	201:1.4 (D + F	-0.04	-0.03	**0.00**	0.00	0.00	-3.77	-2.54	-2.47
Max Sy	126	201:1.4 (D + F	-0.04	-0.03	0.00	**0.00**	0.00	-3.77	-2.54	-2.47
Min Sy	126	201:1.4 (D + F	-0.04	-0.03	0.00	**0.00**	0.00	-3.77	-2.54	-2.47
Max Sxy	126	201:1.4 (D + F	-0.04	-0.03	0.00	0.00	**0.00**	-3.77	-2.54	-2.47
Min Sxy	126	201:1.4 (D + F	-0.04	-0.03	0.00	0.00	**0.00**	-3.77	-2.54	-2.47
Max Mx	191	206:1.4 (D + F	0.10	0.01	0.00	0.00	0.00	**19.01**	-6.79	1.77
Min Mx	142	206:1.4 (D + F	0.16	-0.08	0.00	0.00	0.00	**-32.53**	-14.37	8.58
Max My	155	206:1.4 (D + F	-0.00	-0.11	0.00	0.00	0.00	-2.77	**28.43**	1.29
Min My	133	206:1.4 (D + F	0.04	-0.17	0.00	0.00	0.00	-10.90	**-35.93**	3.44
Max Mxy	234	206:1.4 (D + F	0.05	0.10	0.00	0.00	0.00	6.19	19.31	**14.13**
Min Mxy	343	206:1.4 (D + F	-0.05	0.10	0.00	0.00	0.00	6.19	19.31	**-14.13**

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	Part	
Job Title Mansourah Massarah Gold Project	Ref	
	By AvG / Date 12-Feb-20 / Chd SJM	
Client Ma'aden	File POX CIL (3610).std	Date/Time 25-Feb-2020 10:45

Plate Center Stress Summary

			Shear		Membrane			Bending		
	Plate	L/C	Qx (N/mm^2)	Qy (N/mm^2)	Sx (N/mm^2)	Sy (N/mm^2)	Sxy (N/mm^2)	Mx (kN^-m/m)	My (kN^-m/m)	Mxy (kN^-m/m)
Max Qx	28	206:1.4 (D + F	**0.01**	0.16	-0.07	-0.08	-0.02	-9.62	-56.45	-1.29
Min Qx	25	206:1.4 (D + F	**-0.01**	0.16	-0.07	-0.08	0.02	-9.62	-56.45	1.29
Max Qy	26	206:1.4 (D + F	-0.00	**0.17**	-0.07	-0.10	0.01	-10.39	-59.05	0.39
Min Qy	98	202:0.9 D + 1.6	-0.00	**-0.01**	-0.01	-0.00	-0.00	-0.01	0.88	0.03
Max Sx	116	204:1.2 (D + F	-0.00	0.02	**0.10**	-0.07	0.00	-1.39	-4.05	-0.06
Min Sx	30	206:1.4 (D + F	-0.00	0.15	**-0.07**	-0.05	0.01	-8.70	-53.19	0.46
Max Sy	98	205:0.9 (D + F	-0.00	0.01	0.02	**-0.00**	-0.00	-0.72	-1.04	0.03
Min Sy	44	204:1.2 (D + F	-0.00	0.14	-0.06	**-0.12**	0.01	-8.63	-48.96	0.34
Max Sxy	37	204:1.2 (D + F	-0.01	0.12	-0.05	-0.08	**0.03**	-7.49	-43.96	0.64
Min Sxy	52	204:1.2 (D + F	0.01	0.12	-0.05	-0.08	**-0.03**	-7.49	-43.96	-0.64
Max Mx	120	204:1.2 (D + F	-0.00	0.02	0.08	-0.06	-0.01	**0.30**	-3.80	0.24
Min Mx	26	206:1.4 (D + F	-0.00	0.17	-0.07	-0.10	0.01	**-10.39**	-59.05	0.39
Max My	62	202:0.9 D + 1.6	-0.00	-0.00	0.00	-0.01	-0.00	0.22	**1.90**	0.04
Min My	26	206:1.4 (D + F	-0.00	0.17	-0.07	-0.10	0.01	-10.39	**-59.05**	0.39
Max Mxy	25	206:1.4 (D + F	-0.01	0.16	-0.07	-0.08	0.02	-9.62	-56.45	**1.29**
Min Mxy	28	206:1.4 (D + F	0.01	0.16	-0.07	-0.08	-0.02	-9.62	-56.45	**-1.29**

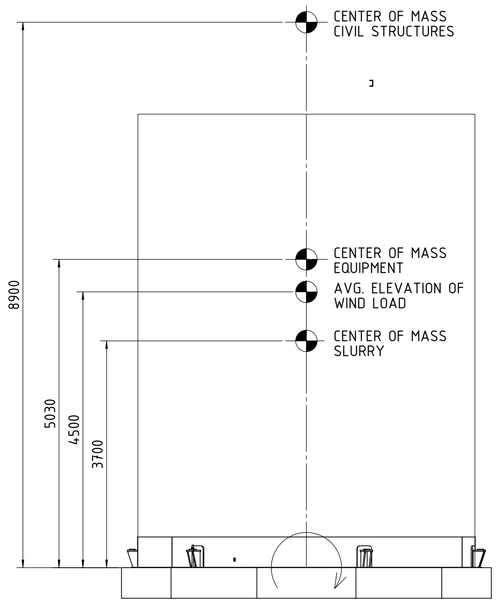
7. REFERENCE

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

ATTACHMENT-1 : LOAD DATA DRAWINGS

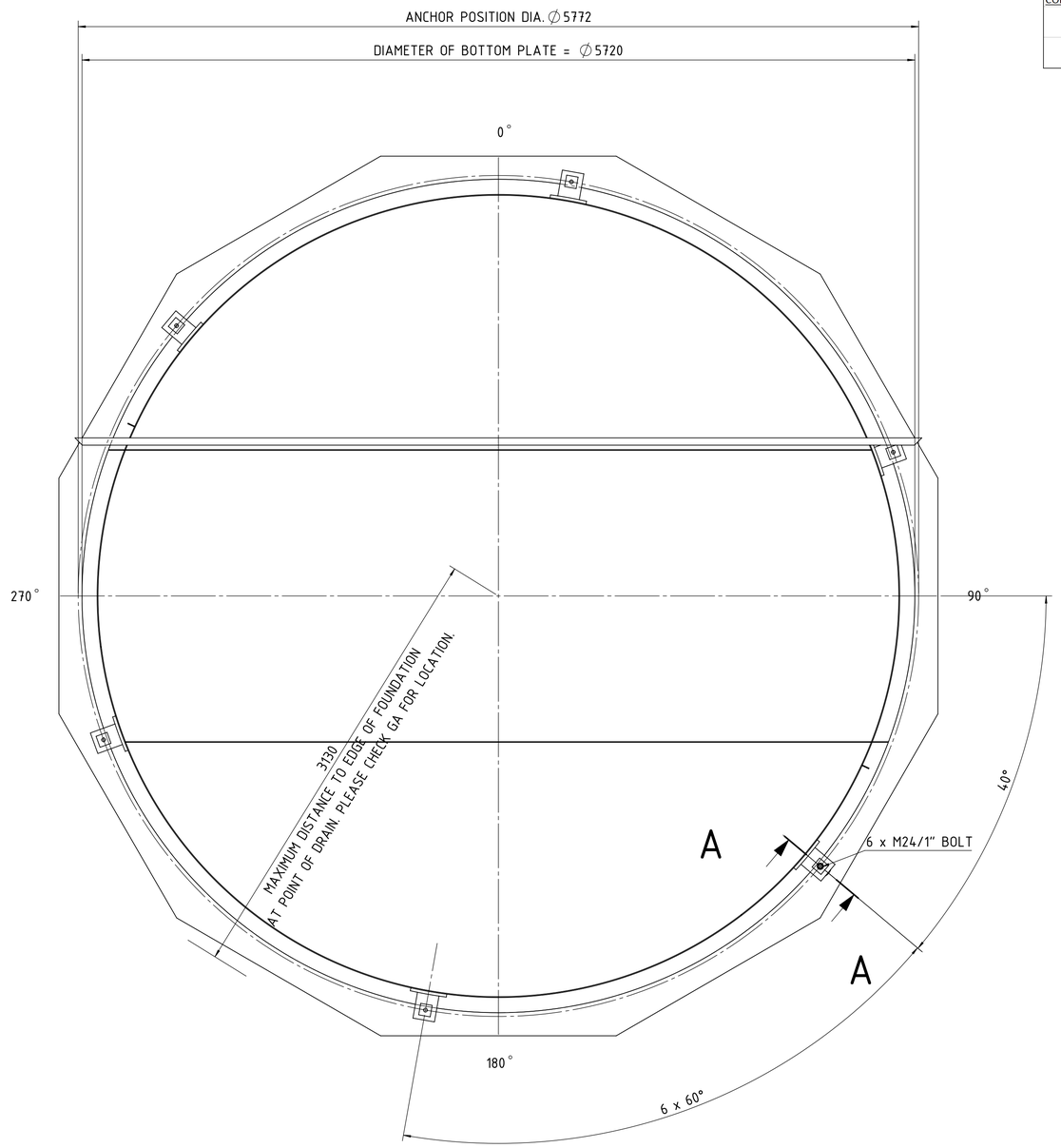
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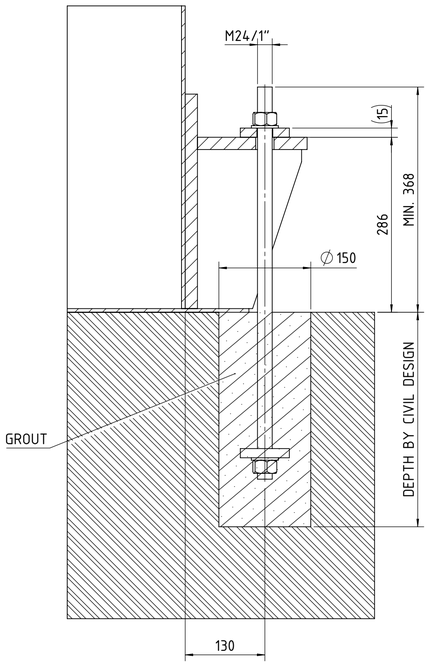
DRAWING NUMBER: MD-216-3610-EG-ME-PLN-1003
REV: A01



SCALE 1 : 50

LOADING NOTES:	
1	THE LOWEST NATURAL FREQUENCY OF COMBINED MACHINE, FLUID LOAD, AND SUPPORT SYSTEM IN ANY DIRECTION MUST EXCEED 1.5 TIMES THE ROTOR ROTATIONAL FREQUENCY (REFER NORMAL OPERATING SPEED*)
2	ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON THE GROUND OR CONCRETE SLAB)
FOUNDATION AND SUPPORT TOLERANCES	
1.	THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2.	THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL
CONCRETE NOTES: (WHERE CELLS ARE INSTALLED ON CONCRETE AND HOLD DOWN BOLTS ARE USED)	
1.	CONCRETE FOUNDATION OUTLINE SHOWN GIVES NOMINAL EDGE DISTANCE FOR HOLD-DOWN BOLT TO EDGE OF FOUNDATION AT 110mm
2.	FINAL OVERALL CONCRETE FOUNDATION DIMENSION AND REQUIRED HOLD-DOWNBOLT EDGE DISTANCE TO BE DETERMINED BY CLIENT





SECTION A-A
SCALE 1 : 5

POX LEACHING TANK	
DEAD WEIGHT OF EQUIPMENT	17250 kg
DEAD WEIGHT IF CIVIL STRUCTURES	1400 kg
OPERATIONAL WEIGHT OF SLURRY	205300 kg
MAXIMUM WEIGHT OF SLURRY	230200 kg
EXTERNAL PLATFORM LOAD	9650 kg
NORMAL OPERATING SPEED	52,4 rpm
WIND LOAD	48,7 kN
SEISMIC LOAD	98 kN
SEISMIC TORQUE AT FOUNDATION LEVEL	373 kNm

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
-S1=0.02g
-Ss=0,10g
-max. groud elevation 3m
CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

BECHTEL

Document Number
MD-216-3610-EG-ME-PLN-1003

Revision A01

Approved

Mr Doron Golan - Bechtel Global

Jan 17, 2020, 10:29 PM GMT+2:00

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DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3610-EG-ME-GAD-1011, -1013,-1015,-1017	GENERAL ARRANGEMENT DRAWING								
MD-216-3610-EG-ME-DAT-1006	EQUIPMENT DATASHEET								
MD-216-3610-EG-PR-PID-1011, -1013,-1015,-1017	P&I DIAGRAM								
MD-216-3610-EG-PR-PFD-1002,-1003	PROCESS FLOW DIAGRAM	A01	19.12.2019	ISSUED FOR APPROVAL	JJ	TM	JR	GR	

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LARSEN & TOUBRO DRAWING NUMBER	019045-450-M-D-02920-003		REVISION 1
OUTOTEC DRAWING NUMBER	900707 GAU01 HL12 MLD15 36103	DOCUMENT ID OU601267749	REVISION 1

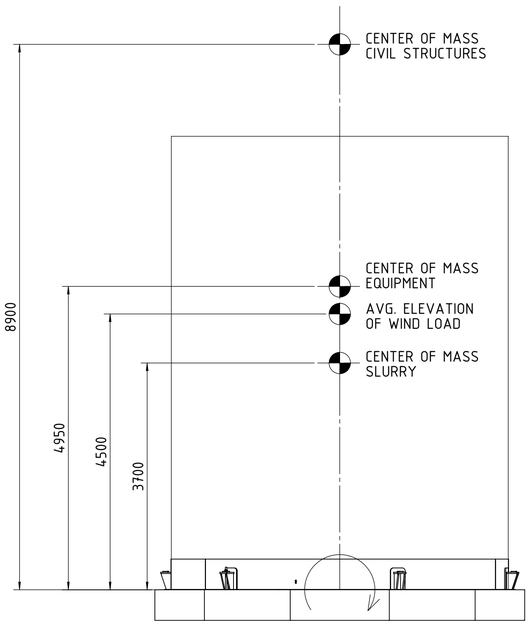
DRAWN	joujaa	19.12.2019
CHECKED	tuoman	19.12.2019
ENGINEER	juhris	19.12.2019
APPROVED	GeoRie	19.12.2019
EMPLOYER		
SCALE	1:20	A1

MANSOURAH - MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
POX CODITIONING, LEACHING AND ADSORPTION
3610-REA-003,005,007,009 POX LEACHING / CIL REACTOR

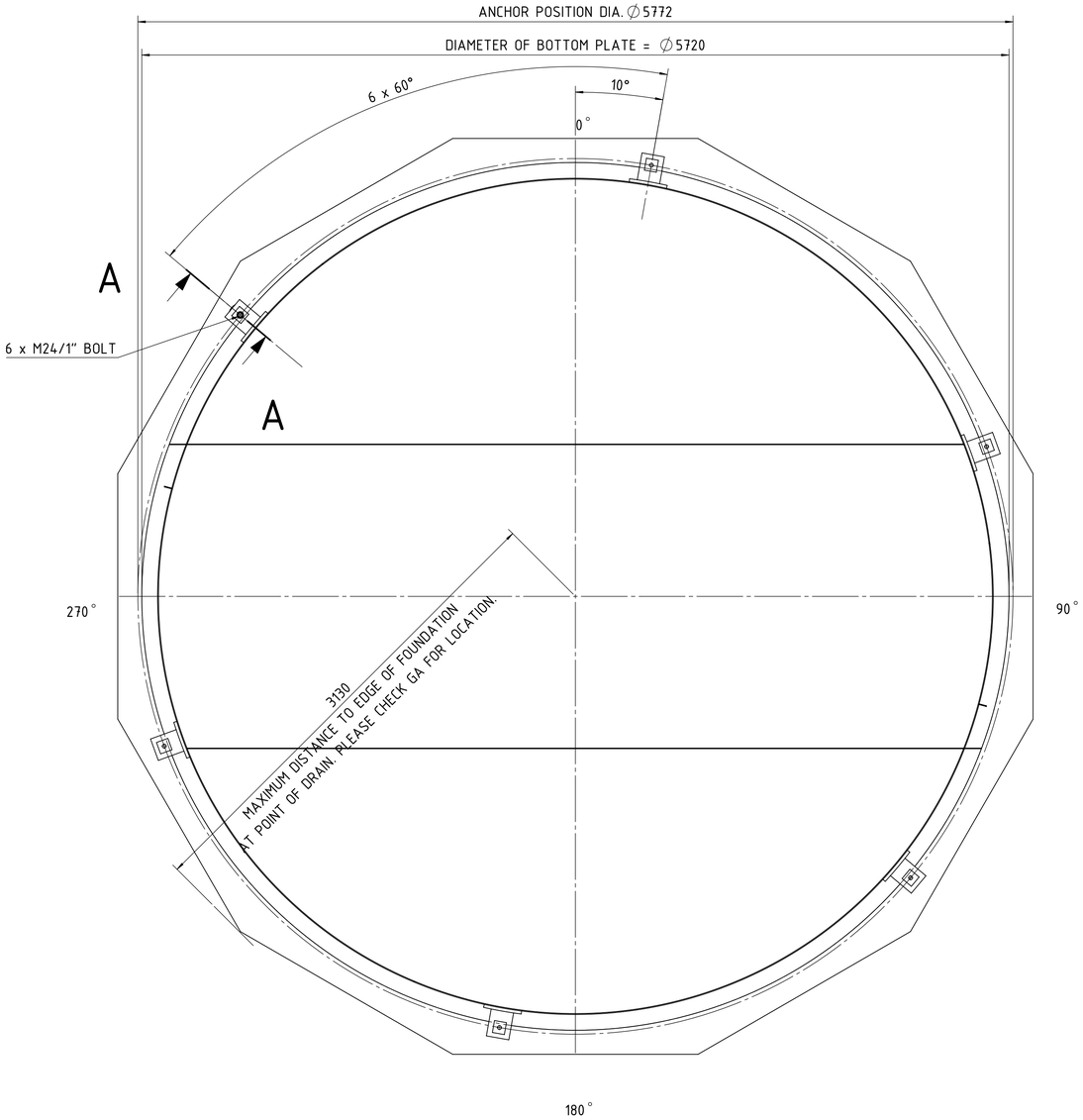
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REV: A01

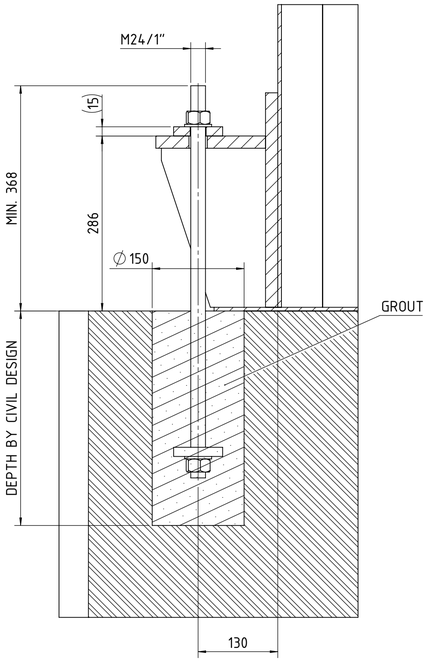
DRAWING NUMBER	REV
MD-216-3610-EG-ME-PLN-1004	A01

LOADING NOTES:	
1	THE LOWEST NATURAL FREQUENCY OF COMBINED MACHINE, FLUID LOAD, AND SUPPORT SYSTEM IN ANY DIRECTION MUST EXCEED 1.5 TIMES THE ROTOR ROTATIONAL FREQUENCY (REFER NORMAL OPERATING SPEED*)
2	ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON THE GROUND OR CONCRETE SLAB)
FOUNDATION AND SUPPORT TOLERANCES	
1.	THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2.	THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL
CONCRETE NOTES: (WHERE CELLS ARE INSTALLED ON CONCRETE AND HOLD DOWN BOLTS ARE USED)	
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2.	FINAL OVERALL CONCRETE FOUNDATION DIMENSION AND REQUIRED HOLD-DOWNBOLT EDGE DISTANCE TO BE DETERMINED BY CLIENT



SCALE 1 : 50





SECTION A-A
SCALE 1 : 5

POX CIL TANK	
DEAD WEIGHT OF EQUIPMENT	17250 kg
DEAD WEIGHT IF CIVIL STRUCTURES	1400 kg
OPERATIONAL WEIGHT OF SLURRY	205300 kg
MAXIMUM WEIGHT OF SLURRY	230200 kg
EXTERNAL PLATFORM LOAD	9650 kg
NORMAL OPERATING SPEED	52,4 rpm
WIND LOAD	98 kN
SEISMIC TORQUE AT FOUNDATION LEVEL	373 kNm

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
-S1=0.02g
-Ss=0,10g
-max. groud elevation 3m
CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

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Document Number
MD-216-3610-EG-ME-PLN-1004

Revision A01

Approved

Mr Doron Golan - Bechtel Global

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DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3610-EG-ME-GAD-1012, -1014, -1016, -1019	GENERAL ARRANGEMENT DRAWING
MD-216-3610-EG-ME-DAT-1006	EQUIPMENT DATASHEET
MD-216-3610-EG-PR-PID-1012, -1014, -1016, -1018	P&I DIAGRAM
MD-216-3610-EG-PR-PFD-1002, -1003	PROCESS FLOW DIAGRAM

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	19.12.2019	ISSUED FOR APPROVAL	JJ	TM	JR	GR	

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LARSEN & TOUBRO DRAWING NUMBER: 019045-450-M-D-02920-004 — REVISION: 0

OUTOTEC DRAWING NUMBER: 900707 GAU01 HL12 MLD15 36104 — DOCUMENT ID: OU601268411 — REVISION: 1

DRAWN	joujaa	19.12.2019
CHECKED	tuoman	19.12.2019
ENGINEER	juhris	19.12.2019
APPROVED	GeoRie	19.12.2019
EMPLOYER		
SCALE	1:20	A1

MANSOURAH - MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
POX CONDITIONING, LEACHING AND ADSORPTION
3610-REA-004, 006, 008, 010 POX CIL REACTOR

DRAWING NUMBER: MD-216-3610-EG-ME-PLN-1004 — REV: A01

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MD-216-3610-EG-CV-CAL-1304	Mansourah Massarah Gold Project	
900707_GAU01_HL12_CED01_36102 _OU100672892_R3	DESIGN CALCULATION POX CONDITIONING, LEACHING & ADSORPTION DESIGN OF TANK FOUNDATION	Date: 22.11.2023
O19045_450_C_K_24131_001_R3	AS BUILT	Rev: D01

ATTACHMENT-2 : STAAD OUTPUT

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     3. ENGINEER DATE 12-FEB-20
     4. JOB NAME MANSOURAH MASSARAH GOLD PROJECT
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045
     7. ENGINEER NAME AVG
     8. CHECKER NAME SJM
     9. APPROVED NAME AG
    10. END JOB INFORMATION
    11. INPUT WIDTH 79
    12. UNIT METER KN
    13. JOINT COORDINATES
    14. 3 0 0 4.027; 4 -3.487 0 2.013; 5 -3.487 0 0; 6 -3.487 0 -2.013; 7 0 0 -4.027
    15. 8 3.487 0 -2.013; 9 3.487 0 0; 10 3.487 0 2.013; 11 0 0 0; 12 -2.75 0 0
    16. 13 -2.70822 0 0.477532; 14 -2.58415 0 0.940555; 15 -2.38157 0 1.375
    17. 16 -2.10662 0 1.76767; 17 -1.76767 0 2.10662; 18 -1.375 0 2.38157
    18. 19 -0.940555 0 2.58415; 20 -0.477532 0 2.70822; 21 0 0 2.75
    19. 22 0.477532 0 2.70822; 23 0.940555 0 2.58415; 24 1.375 0 2.38157
    20. 25 1.76767 0 2.10662; 26 2.10662 0 1.76767; 27 2.38157 0 1.375
    21. 28 2.58415 0 0.940555; 29 2.70822 0 0.477532; 30 2.75 0 0
    22. 31 2.70822 0 -0.477532; 32 2.58415 0 -0.940555; 33 2.38157 0 -1.375
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    25. 40 -0.477532 0 -2.70822; 41 -0.940555 0 -2.58415; 42 -1.375 0 -2.38157
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    32. 57 -1.68383E-016 0.516667 2.75; 58 0.477532 0.516667 2.70822
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233. START GROUP DEFINITION
234. ELEMENT
235. _WALL 18 TO 125
236. _RAFT_OUTER 126 TO 147 154 156 157 161 192 193 200 201 208 TO 213 227 TO 233 -
237. 235 TO 243 256 TO 259 272 TO 275 288 TO 291 293 294 299 TO 306 310 311 315 -
238. 316 320 321 325 326 330 331 335 336 344 TO 353
239. _RAFT_INNER 148 TO 153 155 158 TO 160 162 TO 191 194 TO 199 202 TO 207 214 -
240. 215 TO 226 234 244 TO 255 260 TO 271 276 TO 287 292 295 TO 298 307 TO 309 -
241. 312 TO 314 317 TO 319 322 TO 324 327 TO 329 332 TO 334 337 TO 343
242. END GROUP DEFINITION
243. ELEMENT PROPERTY
244. 126 TO 353 THICKNESS 0.5
245. 18 TO 125 THICKNESS 0.6
246. DEFINE MATERIAL START
247. ISOTROPIC CONCRETE
248. E 2.17185E+007
249. POISSON 0.17
250. DENSITY 23.5616
251. ALPHA 1E-005
252. DAMP 0.05
253. TYPE CONCRETE
254. STRENGTH FCU 27579
255. END DEFINE MATERIAL
256. CONSTANTS
257. MATERIAL CONCRETE ALL
258. SUPPORTS
259. 126 TO 353 PLATE MAT DIRECT Y SUBGRADE 40500
260. LOAD 1 LOADTYPE DEAD  TITLE DEAD LOAD ( D )
261. SELFWEIGHT Y -1
262. JOINT LOAD
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263. 120 TO 155 FY -7.861
264. ELEMENT LOAD
265. 148 TO 153 155 158 TO 160 162 TO 191 194 TO 199 202 TO 207 214 TO 226 234 -
266. 244 TO 255 260 TO 271 276 TO 287 292 295 TO 298 307 TO 309 312 TO 314 317 -
267. 318 TO 319 322 TO 324 327 TO 329 332 TO 334 337 TO 343 PR GY -26
268. 126 TO 147 154 156 157 161 192 193 200 201 208 TO 213 227 TO 233 235 TO 243 -
269. 256 TO 259 272 TO 275 288 TO 291 293 294 299 TO 306 310 311 315 316 320 321 -
270. 325 326 330 331 335 336 344 TO 353 PR GY -14
271. 18 TRAP JT 13 13 8.66666 8.66666
272. 19 TRAP JT 13 13 8.66666 8.66666
273. 20 TRAP JT 13 13 8.66666 8.66666
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276. 23 TRAP JT 13 13 8.66666 8.66666
277. 24 TRAP JT 13 13 8.66666 8.66666
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288. 35 TRAP JT 13 13 8.66666 8.66666
289. 36 TRAP JT 13 13 8.66666 8.66666
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377. 124 TRAP JT 4.33336 4.33336 0 0
378. 125 TRAP JT 4.33336 4.33336 0 0
379. LOAD 2 LOADTYPE FLUIDS  TITLE OPERATIONAL FLUID LOAD ( F1 )
380. ELEMENT LOAD
381. 19 TRAP JT 43.227 43.227 43.227 43.227
382. 20 TRAP JT 43.227 43.227 43.227 43.227
383. 21 TRAP JT 43.227 43.227 43.227 43.227
384. 22 TRAP JT 43.227 43.227 43.227 43.227
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487. 125 TRAP JT 43.227 43.227 43.227 43.227
488. 18 TRAP JT 43.227 43.227 43.227 43.227
489. 148 TO 153 155 158 TO 160 162 TO 191 194 TO 199 202 TO 207 214 TO 226 234 -
490. 244 TO 255 260 TO 271 276 TO 287 292 295 TO 298 307 TO 309 312 TO 314 317 -
491. 318 TO 319 322 TO 324 327 TO 329 332 TO 334 337 TO 343 PR GY -86.5
492. LOAD 3 LOADTYPE FLUIDS  TITLE MAX FLUID LOAD ( F2 )
493. ELEMENT LOAD
494. 19 TRAP JT 48.47 48.47 48.47 48.47
495. 20 TRAP JT 48.47 48.47 48.47 48.47
496. 21 TRAP JT 48.47 48.47 48.47 48.47
497. 22 TRAP JT 48.47 48.47 48.47 48.47
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599. 124 TRAP JT 48.47 48.47 48.47 48.47
600. 125 TRAP JT 48.47 48.47 48.47 48.47
601. 18 TRAP JT 48.47 48.47 48.47 48.47
602. 148 TO 153 155 158 TO 160 162 TO 191 194 TO 199 202 TO 207 214 TO 226 234 -
603. 244 TO 255 260 TO 271 276 TO 287 292 295 TO 298 307 TO 309 312 TO 314 317 -
604. 318 TO 319 322 TO 324 327 TO 329 332 TO 334 337 TO 343 PR GY -96.9
605. LOAD 4 LOADTYPE WIND  TITLE WIND LOAD ( W )
606. JOINT LOAD
607. 121 FY 0.769
608. 122 FY 1.515
609. 123 FY 2.213
610. 124 FY 2.846
611. 125 FY 3.392
612. 126 FY 3.834
613. 127 FY 4.16
614. 128 FY 4.359
615. 129 FY 4.427
616. 130 FY 4.359
617. 131 FY 4.16
618. 132 FY 3.834
619. 133 FY 3.392
620. 134 FY 2.846
621. 135 FY 2.213
622. 136 FY 1.515
623. 137 FY 0.769
624. 139 FY -0.769
625. 140 FY -1.515
626. 141 FY -2.213
627. 142 FY -2.846
628. 143 FY -3.392
629. 144 FY -3.834
630. 145 FY -4.16
631. 146 FY -4.359
632. 147 FY -4.427
633. 148 FY -4.359
634. 149 FY -4.16
635. 150 FY -3.834
636. 151 FY -3.392
637. 152 FY -2.846
638. 153 FY -2.213
639. 154 FY -1.515
640. 155 FY -0.769
641. 120 TO 155 FZ -1.353
642. LOAD 5 LOADTYPE SEISMIC  TITLE SEISMIC LOAD ( E )
643. JOINT LOAD
644. 121 FY 1.31
645. 122 FY 2.578
646. 123 FY 3.767
647. 124 FY 4.844
648. 125 FY 5.773
649. 126 FY 6.526
650. 127 FY 7.08
651. 128 FY 7.419
652. 129 FY 7.534
653. 130 FY 7.419
654. 131 FY 7.08
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655. 132 FY 6.526
656. 133 FY 5.773
657. 134 FY 4.844
658. 135 FY 3.767
659. 136 FY 2.578
660. 137 FY 1.31
661. 139 FY -1.31
662. 140 FY -2.578
663. 141 FY -3.767
664. 142 FY -4.844
665. 143 FY -5.773
666. 144 FY -6.526
667. 145 FY -7.08
668. 146 FY -7.419
669. 147 FY -7.534
670. 148 FY -7.419
671. 149 FY -7.08
672. 150 FY -6.526
673. 151 FY -5.773
674. 152 FY -4.844
675. 153 FY -3.767
676. 154 FY -2.578
677. 155 FY -1.31
678. 120 TO 155 FZ -2.722
679. LOAD COMB 201 1.4 ( D + F1 )
680. 1 1.4 2 1.4
681. LOAD COMB 202 0.9 D + 1.6 W
682. 1 0.9 4 1.6
683. LOAD COMB 203 1.2 ( D + F1 ) + 1.6 W
684. 1 1.2 2 1.2 4 1.6
685. LOAD COMB 204 1.2 ( D + F1 ) + E
686. 1 1.2 2 1.2 5 1.0
687. LOAD COMB 205 0.9 ( D + F1 ) + E
688. 1 0.9 2 0.9 5 1.0
689. LOAD COMB 206 1.4 ( D + F2 )
690. 1 1.4 3 1.4
691. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**351**	**NUMBER OF MEMBERS**	**0**
NUMBER OF PLATES	**336**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**243**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 341/ 33/ 135 DOF
TOTAL PRIMARY LOAD CASES = 5, TOTAL DEGREES OF FREEDOM = 1377
TOTAL LOAD COMBINATION CASES = 6 SO FAR.
SIZE OF STIFFNESS MATRIX = 186 DOUBLE KILO-WORDS

DXF IMPORT OF 3610_NEW.DXF **-- PAGE NO. 14**

REQRD/AVAIL. DISK SPACE = 15.8/ 51458.1 MB

692. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= FEB 25,2020 TIME= 10:45:42 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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